

Special lab #2: Single photon interference

1/ The difference between waves and particles is that waves interfere and exhibit diffraction patterns. They have amplitudes, wavelengths (and frequencies). However, particles are classically understood as smallest portions of matter characterized by mass, volume, density (atoms, molecules)

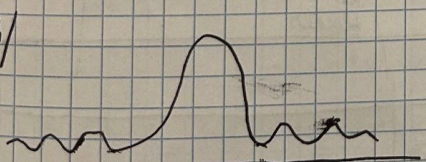
Wave-particle duality is when a particle behaves like a wave (for example: interferes) and when waves behave like particles (they're called wave quanta) and their energy is distributed in packets.

2/ Single slit formula: $b \sin \theta_{m0} = m_0 \lambda$ } Lect chapter 10

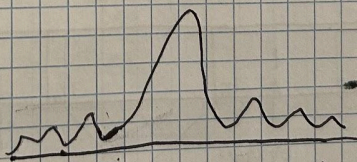
Double slit formula: $a \sin \theta_{mI} = mI \lambda$

b : wideness, a : slit separation, $mI = \pm \text{# subsidiary peaks}$

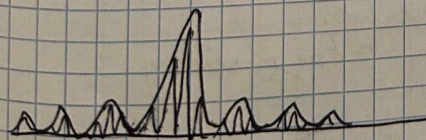
3/



a/ centered in front of left slit

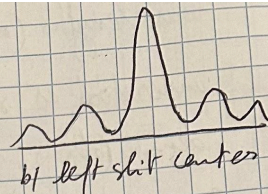
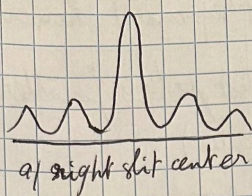
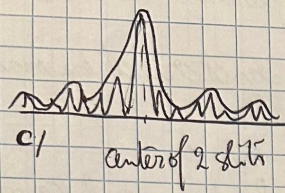


b/ centered in front of right slit



4/ Diffraction + interference

4/



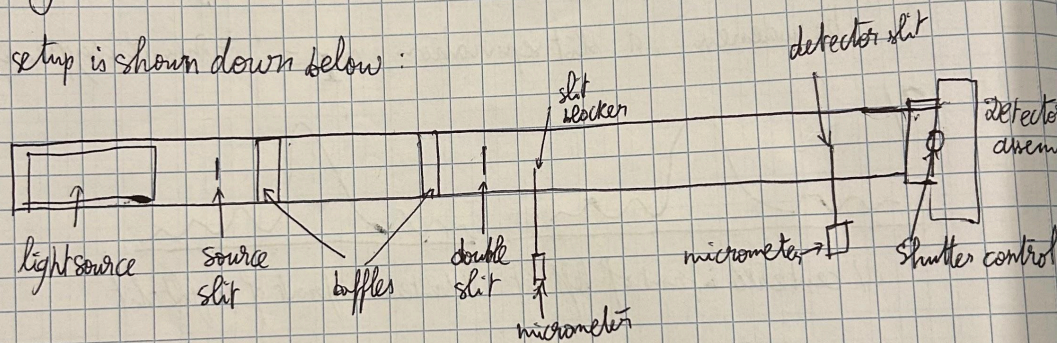
Experiment:

light source: laser, light bulb

detectors: Photodiode, Photomultiplier Tube
 ↑ less sensitive ↑ more sensitive

Moving the shutter allows us to select the detector

The setup is shown down below:



Safety rules: not to look directly into the laser
 not to exceed 900 V voltage setting

- Measure: distance from double slits to detector slit
- switch to light bulb and PMT
- open slit blocker
- detector slit at central maximum
- laser off, remove cover of channel
- lift light bulb in position (+change brightness)
- Cover on, shutter closed, PMT high voltage off, 10 turn dial at 0
- light intensity at half maximum
- PMT output $\rightarrow$ oscilloscope [Channel 1]
 - $\hookrightarrow$ input of pulse counter

- output of pulse counter $\rightarrow$ oscilloscope [Channel 2]
- 0.5V pulse for each input pulse exceeding discriminator threshold

variables: $\left\{ \begin{array}{l} \text{PMT voltage} \\ \text{Amp gain} \\ \text{Discrimination threshold} \end{array} \right\}$ $\left\{ \begin{array}{l} \text{1 turn} \rightarrow 0.1V \\ \text{count real photons} \\ \text{not electronic noise} \end{array} \right\}$

- PMT on, ramp voltage up, monitor w/ voltmeter (DC - 1V range)
 - $\hookrightarrow \frac{1}{1000}$ of PMT voltage

inc voltage to 850V (0.85V on voltmeter)

- Trigger oscilloscope (on channel 1): PMT signal
- Trigger oscilloscope (on channel 2): discriminator
- length of gate / count time: 10s

Gate toggle switch $\rightarrow$ Manual: single measurement
 $\hookrightarrow$ Auto: repeated (rep rate)

- Note dark rate of PMT

- Bulb intensity : PMT count rate on central max (2 slit mode)
shutter open < 50000 c/s
5% efficient (if rate is 50000 c/s $\rightarrow$ tot rate = 20×50000 c/s)
- Measure rate of photons in 10s intervals as the detector slit is moved w/ both double and single slits.
- locate detector slit to get max/min count rate
for 2 slits, move slit blocker : observe for 1 slit vs 2 slits

Lab Notes:

micrometer
slits: 6.25 mm

1 slit (left): 5.12 mm ± 0.005
exposed

1 slit (right): 7.08 mm ± 0.005
exposed

2 blocked : 4.75 mm ± 0.005 mm
8.18 mm ± 0.005 mm

center max: 4.34 mm, 1.925 V (both slots)

both blocked: 4.5 mm, 0.009 V

left open: 5.12 mm, 0.573 V

both open: 6.25 mm, 1.927 V

right open: 7.08 mm, 0.431 V

adjacent minimum 1: 3.96 mm, 0.105 V (both slots)

both blocked: 4.5 mm, 0.009 V

left open: 5.12 mm, 0.615 V

both open: 6.25 mm, 0.105 V

right open: 7.08 mm, 0.390 V

adjacent min 2: 4.69 mm, 0.081 V

both blocked: 4.5 mm, 0.009 V

left slit: 5.12 mm, 0.503 V

both open: 6.25, 0.081 V

right open: 7.08, 0.456 V

Single slit position (mm)	Single photon Counts	position (mm)	Counts
1.0	564	5.2	3821
1.2	870	5.4	3087
1.4	1204	5.6	2443
1.6	1598	5.8	2044
1.8	2139	6.0	1429
2.0	2642	6.2	1136
2.2	3261	6.4	807
2.4	3966	6.6	563
2.6	4577	6.8	466
2.8	5148	7.0	365
3.0	5682	7.2	352
3.2	6095	7.4	339
3.4	6637	7.6	404
3.6	6712	7.8	480
3.8	6759	8.0	504
4.0	6769		
4.2	6447		
4.4	6082		
4.6	5642		
4.8	4971		
5.0	4342		

Wuh

Setup:

Figure 1: U tube picture

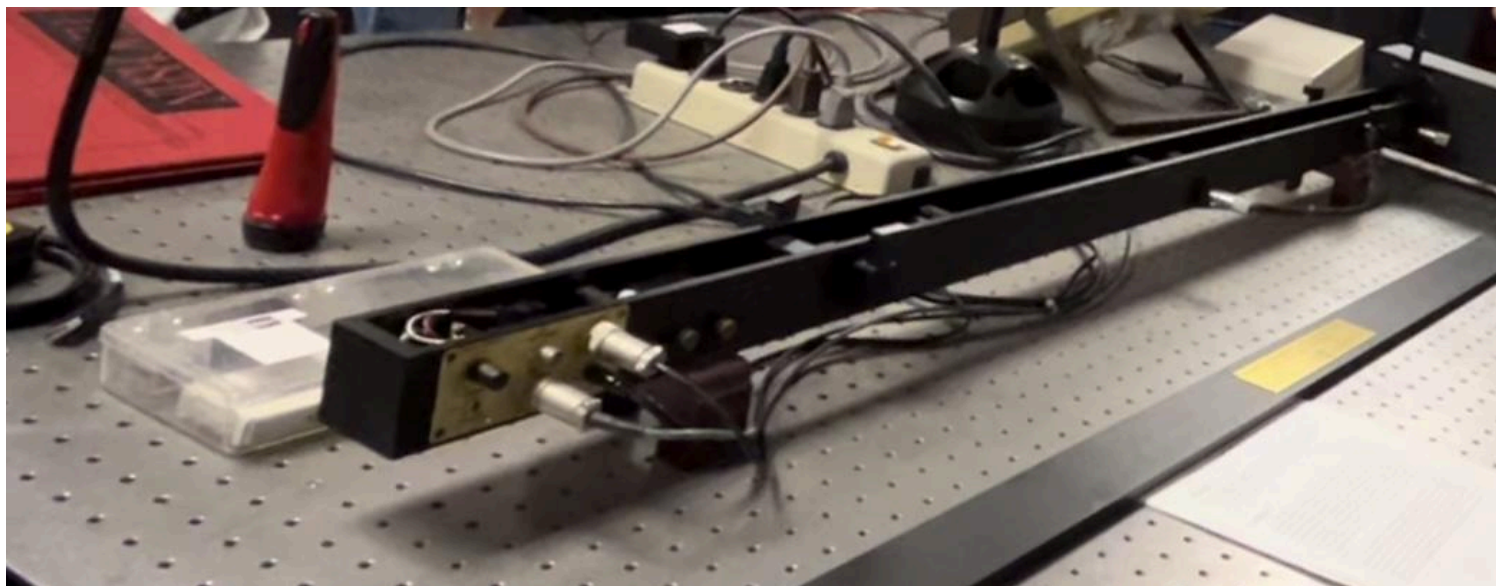


Figure 2: laser and lamp control



Figure 3: Control



Figure 4: PMT control



Figure 5: Counter



Figure 6: Oscilloscope showing PMT output (yellow) and counter output (blue)

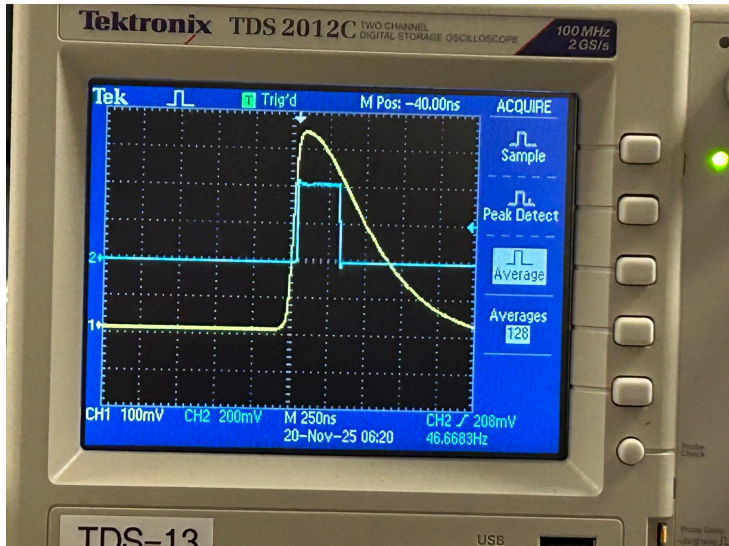


Figure 7: Single slit diffraction

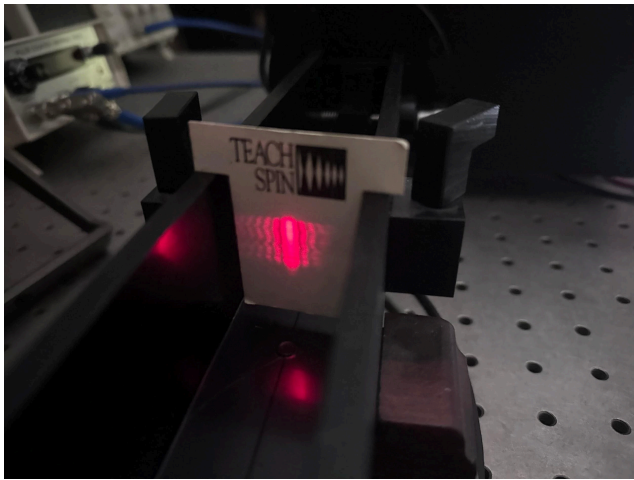
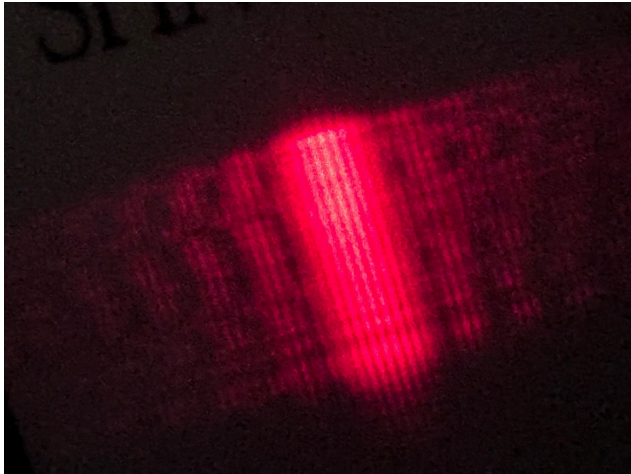


Figure 8: Double slit interference



Procedure:

Before powering up the system, we made sure the PMT high voltage is off (on the front panel of the detector unit), the shutter is closed, and the power switch for the light source is in the OFF position (far end of the U channel).

With the cover of the U channel still removed, we plug in the power adapter and turn the laser on.

We check the alignment of the system by following the light beam as it passes through the single and then the double slit.

We place a white card in front of the laser beam and move it down the U channel to observe the change in the diffraction/interference pattern.

The card placed just behind the single slit has a single thin light strip. As we move the card downstream we see the single slit diffraction pattern appear and spread out (see figure 7).

The middle bright fringe falls right on the double slit.

Placing the card behind the double slit/slit blocker assembly shows a pair of thin red lines of laser light. As we move the card farther downstream, the characteristic double slit interference pattern is shown on the card (see Figure 8) just in front of the detector slit.

When we block one of the two slits by turning the micrometer attached to the slit blocker just behind the double slit, the interference pattern disappears and we only observe diffraction (figure 7).

We move the card back just behind the double slit/slit blocker and record the position of the micrometer readings of the slit blocker for both slits open, only left slit open, only right slit open, and both slits blocked (see table 1)

Table 1:

Slit state	Position (mm)
2 slits open	6.25
left slit open	5.12
right slit open	7.08
2 blocked	4.75
2 blocked	8.18

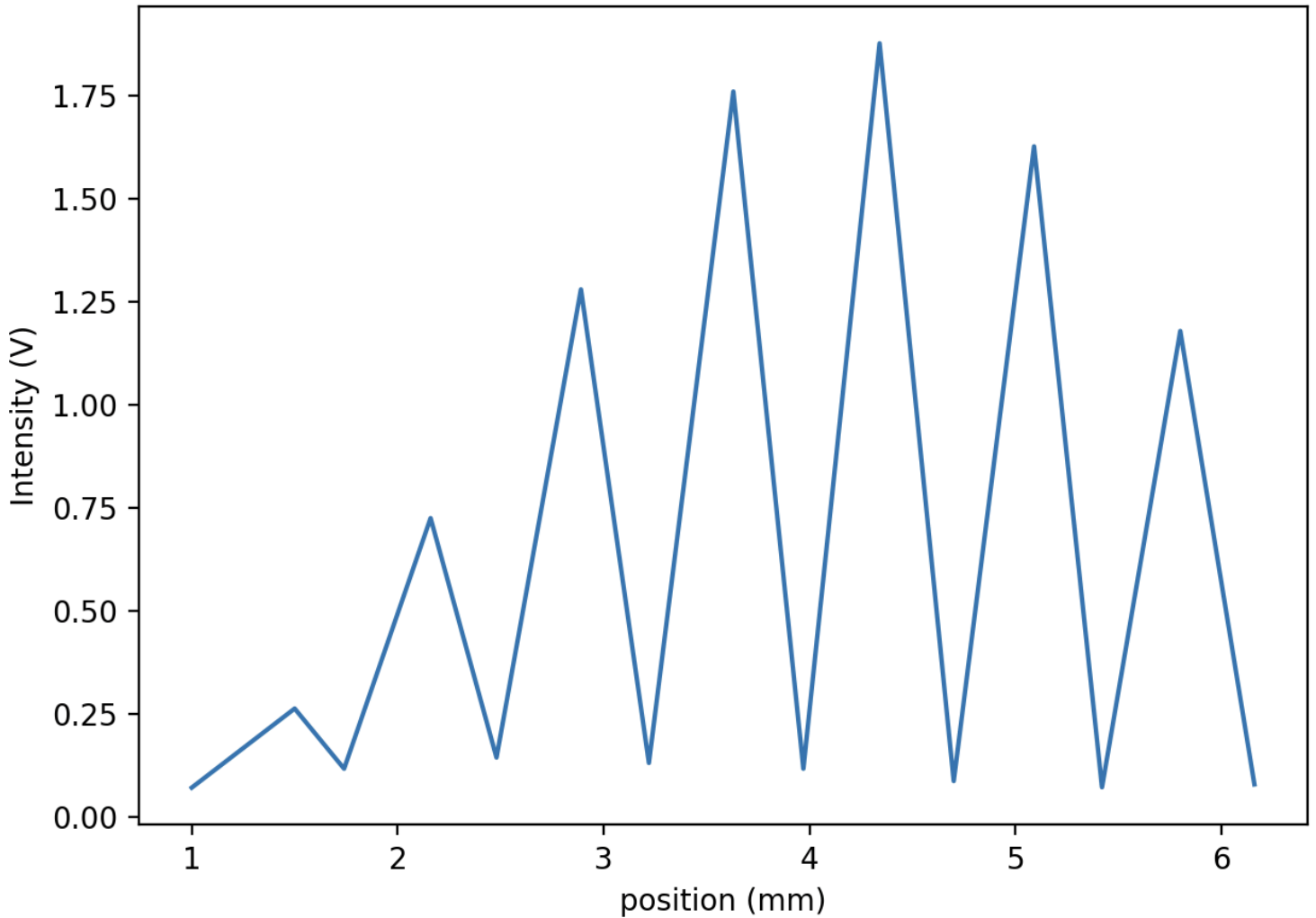
We put the cover of the U channel back in place and turn on the digital voltmeter which is connected to the photodiode output. With the slit blocker in open position for both slits, we move the detector slit with its micrometer across the interference pattern and record the positions and intensity of 4-5 bright (maximum) and dark (minimum) fringes (see table 2)

Table 2: Testing Double Slit

Position (mm)	Intensity (V)
1	0.07
1.5	0.262
1.74	0.116
2.16	0.724
2.48	0.143
2.89	1.279
3.22	0.13
3.63	1.759
3.97	0.116
4.34	1.876
4.7	0.086
5.09	1.626
5.42	0.071
5.8	1.178
6.16	0.078

We plot the data just to make sure that we are getting interference pattern:

LASER PATTERN



- We find the central maximum (the highest intensity at 6.25 mm)
- Then we observe how the light intensity of the central maximum and adjacent minima change in the five settings of the slit blocker (see table 3)

Table 3:

	Central max	position = 4.34 mm	adjacent min 1	position = 3.96 mm	adjacent min 2	position = 4.69 mm
	positions (mm)	Intensity (V)	positions (mm)	Intensity (V)	positions (mm)	Intensity (V)
both slits open	6.25	1.927	6.25	0.105	6.25	0.081
only left slit open	5.12	0.573	5.12	0.615	5.12	0.503
only right slit open	7.08	0.431	7.08	0.39	7.08	0.456
both slits blocked	4.5	0.009	4.5	0.009	4.5	0.009

- These observations tell us that we are getting the central max highest intensity at the position when both slits are open, We also see that the intensity drops by a factor of 4 for single slit versus double slit and that is because there is interference with the double slit, therefore, the wave amplitudes add up, and the intensity is amplitude squared so we get $I(\text{double}) = (E(\text{double}))^2 = (2 * E(\text{single}))^2 = 4 * I(\text{single})$
- Another thing that we have noticed is that the data is not symmetric and that is due to some misalignment, which will not ruin the experiment but will display asymmetric patterns.
- We measure the light intensity as a function of the detector-slit position as it is moved across the front of the photodiode to get a quantitative record of the interference/diffraction pattern for both double slit (table 4) and single slit (table 5)

Table 4: Double Slit Laser Interference Position and Intensity

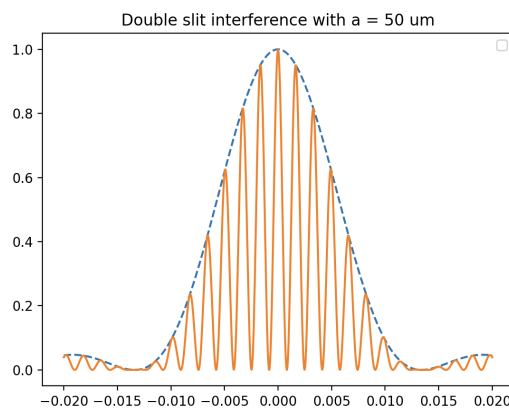
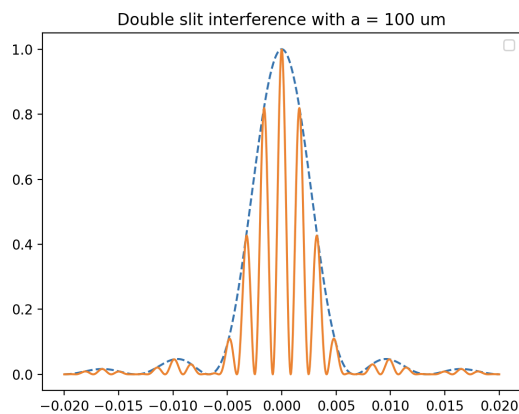
Position (mm)	Intensity (V)
1	0.074
1.1	0.094
1.2	0.146
1.3	0.215
1.4	0.273
1.5	0.267
1.6	0.191
1.7	0.123
1.8	0.151
1.9	0.315
2	0.55
2.1	0.709
2.2	0.685
2.3	0.445
2.4	0.21
2.5	0.16
2.6	0.432
2.7	0.905
2.8	1.263
2.9	1.261
3	0.887
3.1	0.381
3.2	0.134
3.3	0.318
3.4	0.871
3.5	1.468
3.6	1.797
3.7	1.457
3.8	0.763
3.9	0.188
4	0.182
4.1	0.851
4.2	1.583
4.3	1.929
4.4	1.623
4.5	0.9
4.6	0.26
4.7	0.096
4.8	0.545
4.9	1.234
5	1.161
5.1	1.683
5.2	1.574
5.3	0.996
5.4	0.359
5.5	0.07
5.6	0.277
5.7	0.755
5.8	1.148
5.9	1.179
6	0.914
6.1	0.389
6.2	0.103
6.3	0.12
6.4	0.356
6.5	0.584
6.6	0.255

Table 5: Single Slit Laser Diffraction Position and Intensity

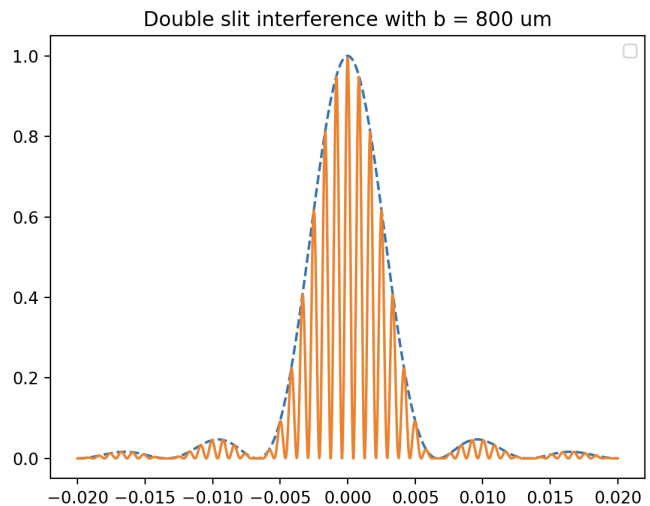
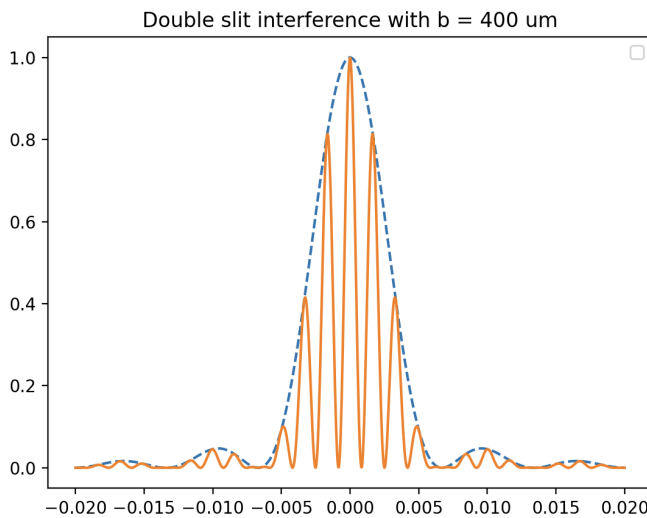
Position (mm)	Intensity (V)
1	0.087
1.2	0.128
1.4	0.172
1.6	0.222
1.8	0.274
2	0.326
2.2	0.382
2.4	0.434
2.6	0.483
2.8	0.527
3	0.563
3.2	0.589
3.4	0.611
3.6	0.626
3.8	0.627
4	0.615
4.2	0.593
4.4	0.56
4.6	0.522
4.8	0.48
5	0.435
5.2	0.386
5.4	0.333
5.6	0.281
5.8	0.231
6	0.186
6.2	0.143
6.4	0.106
6.6	0.075
6.8	0.05
7	0.031
7.2	0.018
7.4	0.011
7.6	0.01
7.8	0.01
8	0.01
8.2	0.01
8.4	0.01
8.6	0.011
8.8	0.01
9	0.01

- Distance from the double slits to the detector slit (measured) 500 mm

- We plot the data (Figures 9 and 10) and fit it to the expected functional form (Figures 11 and 12)
- Our data points should lie on the following curves:
 $i_1 = I_0 * (\text{sinc}(\alpha/\pi))^2$ (for single slit)
 $i_2 = I_0 * (\text{sinc}(\alpha/\pi) * \sin(N * \beta)/\sin(\beta)/N)^2$ (for N slits: our case: N = 2 for 2 slits)
 Such as: $\sin(\theta) = (x - x_0) / D$
 $\alpha = \pi * a * \sin(\theta) / \lambda$
 $\beta = \pi * b * \sin(\theta) / \lambda$
- The following is an approximate calculation of the wavelength of the light we are using (from chapter 5)
 - Taking the location of the central maximum as defining a zero of angle (4.3mm), when we are at the next-to-central maximum (5.1 mm) and using the distance from the double slit to the screen (500 mm) we are off-axis by angle of $(5.1 - 4.3) \text{ mm} / 500 \text{ mm} = 0.0016$
 - The textbook derivation of the result written as $n = d \sin \theta$: $b \sin \theta = m \lambda$, $a \sin \theta = m \lambda$,
 - $n = 1$ is appropriate for the next-to-central maximum because we are looking at constructive interference and n is the integer representing the local maxima
 - We use our computed value of the angle to compute the ratio λ / d : we get 0.0016
 - The double-slit assembly installed in the apparatus is marked with the hand-written digits 14, 16, or 18. These correspond to values of d, the center-to-center separation of the double slits, of 14, 16, or 18 mils = 0.014", 0.016", or 0.018" = 0.356 mm, 0.406 mm, or 0.457 mm.
 - We use our results to compute the wavelength = $\lambda / d * d = 0.000650 \text{ mm} = 650 \text{ nm}$ The wavelength we computed agrees with the color we see (red) and what you've been told from the specifications: 670 nm (% error = 2.99 %)
- The main independent variable will be the setting of the detector-slit micrometer. An auxiliary independent variable will be which of the five settings of the slit-blocker we are using. The dependent variable will be the PHOTODIODE OUTPUT voltage from the current to voltage converter that is built into our detector unit, which is a 'surrogate' for the local intensity in the interference pattern
 - The influence of the slit width a: the larger the slit width is, the narrower the interference pattern is: please see the following two plots:



- The influence of the slit separation d on the shape of the data: the larger b is, the more interference fringes we see. See the plots below that vary only b:



Part 2:

- We move to the double-slit position and move the detector slit to the central maximum.
- We turn the laser off and remove the cover of the U channel.
- Our professor helps us by lifting the light bulb housing in position.
- There is a very narrow band green filter in front of the bulb.
- We turn the light bulb on and verify that it is working and its brightness changes as we change its intensity.
- We place the cover of the U channel back in place and secure it. The PMT protective system allows the photomultiplier to be powered only when the cover is on and the light bulb is turned on. We keep the shutter, which keeps the light from entering the photomultiplier, still closed down.
- We check that the PMT high voltage switch is off and the 10-turn dial is at 0.
- We set the bulb intensity to about half of the maximum.
- We connect the photomultiplier output to channel 1 of the oscilloscope and the input of the pulse counter, and the monitor output of the pulse counter to channel 2 of the oscilloscope. **The monitor output produces a 0.5 V pulse for each input pulse that exceeds the discriminator threshold.** The discriminator threshold can be set by a **10-turn dial on the pulse counter (one full turn corresponds to 0.1 V)**. It sets the minimum pulse height of any input signal that is counted by the counter. The photomultiplier operating voltage, the amplifier gain (on the detector unit), and the discriminator threshold all affect the signal count. They must be set so that the photomultiplier will optimally count real photons and reject electronic noise.
- We turn the PMT power on with the switch and gradually ramp the high voltage up using the dial on the front of the detector unit. The dial does not indicate the actual voltage and we will have to monitor its value through the test points with a voltmeter. In addition, the voltage measured at the test point is 1/1000 of the actual high voltage applied to the PMT (we use DCV measurement in 1 V range). We increase the PMT high voltage gradually to 850 V (i.e. 0.85 V on the voltmeter). We turn off the light bulb or remove the cover of the U channel while the HV is on (the failsafe mechanism immediately cuts the power to the PMT).
- We observe the PMT signal on the oscilloscope by triggering on the PMT output (channel 1) and changing the trigger level. The signals that are real pulses are those that have intensity higher than the counter signal intensity. And since we trigger on the counter output then the pulses that are seen on the oscilloscope above the trigger threshold are the real photons, but the ones that only appear if we trigger the PMT output are electrical noise.
- We then set the oscilloscope to trigger on the discriminator output of the pulse counter (channel 2) and observe both PMT and monitor signals at the same time. Since we trigger on the discriminator signal, we only see the PMT signals on channel 1 that have satisfied the discriminator's voltage criterion, i.e. whose height is higher than the set threshold
- We try changing the discriminator threshold and see how that affects the PMT pulses. We see that for lower thresholds, we get more counts because the discriminator counts any signal above the threshold (figure 6).

If we set the threshold too low, we get too many counts and that means that we are capturing electronic noise from the PMT.

- Relevant parameters: HV = 850 V, threshold = 9.1
- We set the length of the Gate for 10 s on the selector. Pressing the Gate toggle switch briefly to Manual starts a single measurement. With the Gate toggle set to Auto, the counter repeatedly makes measurements with a wait time between the measurements determined by the Rep Rate dial.
- When we open the shutter, counts are at an average of 544 counts
- When we close the shutter, counts are at an average of 183
- The counting rate >0 when the shutter is closed and there is no light reaching the PMT because the PMT reflects the noise from the electric circuits (since we are operating at room temperature, there will be vibrations) in the detector. This is the so-called dark rate of the PMT.
- We set the bulb intensity so that the PMT count rate on the central maximum in the two-slit mode with the shutter open is less than 50,000 counts/second (we got 34616 counts).
- For the green wavelength of interest, the PMT is about 5% efficient; i.e. it counts only about 1 out of 20 green photons. Therefore, if the PMT count rate is on the order of 50×10^3 photons/second, the total rate at which photons reach the PMT is $20 \times 50 \times 10^3$ or about 10^6 /second. The average time between photon arrivals at the PMT is thus 10^{-6} second (i.e. $\sim 1 \mu\text{s}$ or $\sim 1000 \text{ ns}$). Any individual photon traverses the 1 meter length of the apparatus at a speed of light $c = 3 \times 10^8 \text{ m/s}$ ($= 30 \text{ cm/ns}$), spending a time of about 3 ns, in flight. On average, therefore, there is a photon in flight between the first single slit and the photomultiplier for only about 3 ns out of every 1000 ns. This means that for 99.7% of the time there are “no” photons in flight, while for only 0.3% of the time there is “one” photon in flight in the apparatus. The probability of there being two photons in flight simultaneously is thus less than 1 (even if we consider that the total light intensity after the first slit is larger than the light intensity measured at the peak of the double slit interference pattern). All the interference effects observed in this setup can thus be ascribed to the behavior of individual photons, coming through the single slit one at a time.
- We measure the rate of photons in 10 second intervals as we move the detector slit across the face of the PMT with both single slit (see table 6) and double slit (see table 7) open.

Table 6: Single Slit Single Photon Diffraction Position and Counts

Position (mm)	Counts	Counts - dark rate
1	564	20
1.2	870	326
1.4	1204	660
1.6	1598	1054
1.8	2139	1595
2	2642	2098
2.2	3261	2717
2.4	3966	3422
2.6	4577	4033
2.8	5148	4604
3	5682	5138
3.2	6095	5551
3.4	6637	6093
3.6	6712	6168
3.8	6759	6215
4	6769	6225
4.2	6447	5903
4.4	6082	5538
4.6	5642	5098
4.8	4971	4427
5	4342	3798
5.2	3821	3277
5.4	3087	2543
5.6	2443	1899
5.8	2044	1500
6	1429	885
6.2	1136	592
6.4	807	263
6.6	563	19
6.8	466	78
7	365	179
7.2	352	192

Table 7: Double Slit Single Photon Interference Position and Counts

Position (mm)	Counts	Counts - dark rate
1	1290	746
1.1	1114	570
1.2	802	258
1.3	934	390
1.4	975	431
1.5	1392	848
1.6	1569	1025
1.7	2161	1617
1.8	3112	2568
1.9	3805	3261
2	3851	3307
2.1	2379	1835
2.2	2119	1575
2.3	4355	3811
2.4	8460	7916
2.5	10922	10378
2.6	8984	8440
2.7	4863	4319
2.8	2961	2417
2.9	7409	6865
3	13100	12556
3.1	20600	20056
3.2	18150	17606
3.3	13191	12647
3.4	4566	4022
3.5	7851	7307
3.6	19383	18839
3.7	28626	28082
3.8	28338	27794
3.9	18738	18194
4	7185	6641
4.1	8311	7767
4.2	23430	22886
4.3	35160	34616
4.4	33119	32575
4.5	20755	20211
4.6	9675	9131
4.7	10457	9913
4.8	17987	17443
4.9	32556	32012
5	32588	32044
5.1	22285	21741
5.2	11497	10953
5.3	9790	9246
5.4	17101	16557
5.5	24853	24309
5.6	25478	24934
5.7	18709	18165
5.8	10701	10157
5.9	8220	7676
6	11253	10709
6.1	15395	14851
6.2	15662	15118
6.3	12174	11630
6.4	8484	7940
6.5	5928	5384

- We plot the data as a function of position of the detector slit (see figures 13 and 14 for the raw data and 15 and 16 for the data with the fits)
- This graph agrees with the interference/diffraction patterns observed earlier with the photodiode. As we see diffraction with single slit and interference with double slit.
- The spectral distribution of the photons emitted by the light bulb is different from the laser.
- The wavelength of the bulb is 546 nm (Specifications)

When we are done with the experiment, we gradually ramp down the HV and then turn the light bulb off.

Analysis:

Figure 9: Single Slit Laser Diffraction Pattern

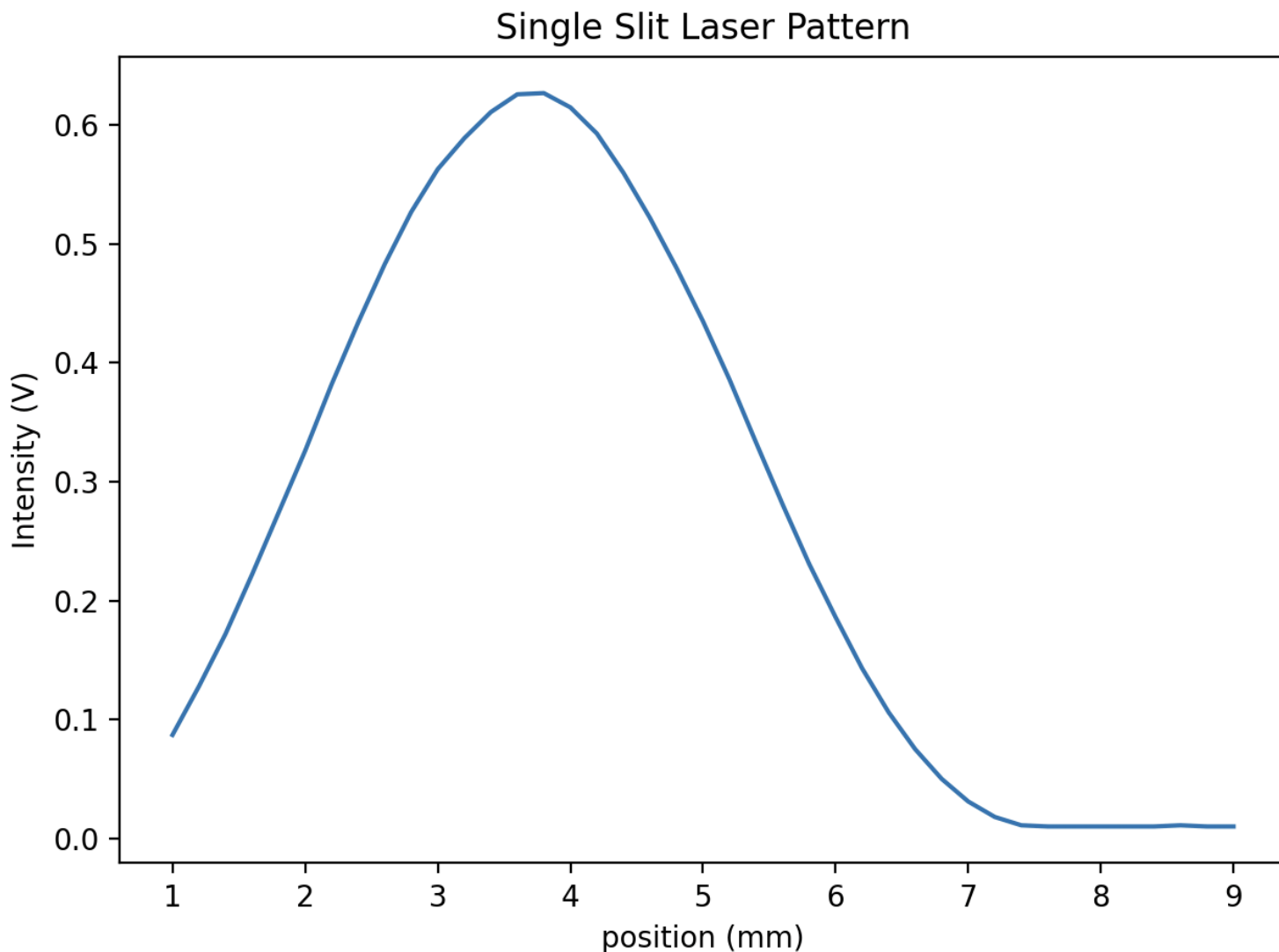


Figure 10: Double Slit Laser Interference Pattern

Double Slit Laser Pattern

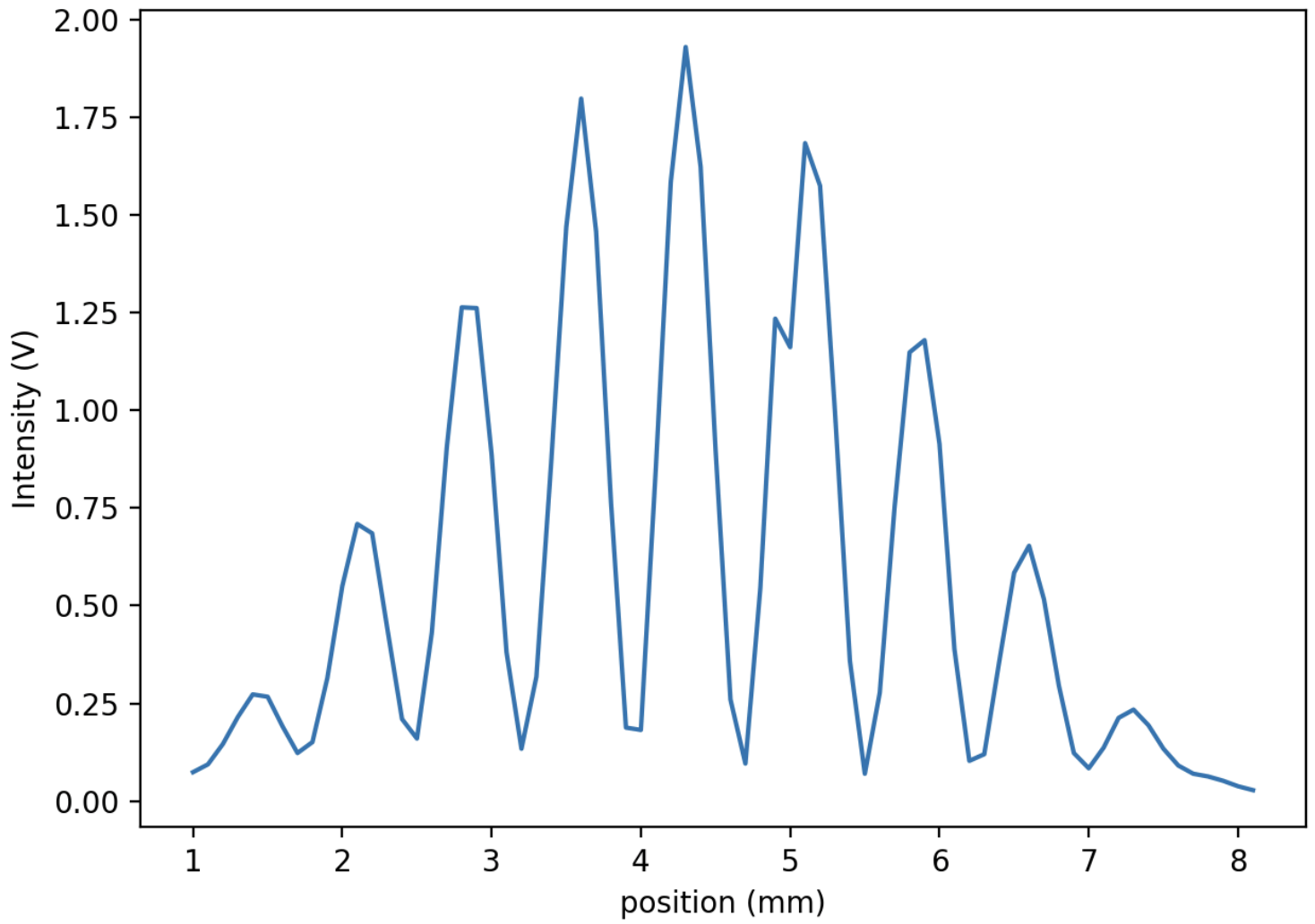


Figure 11 a: Laser single slit fit

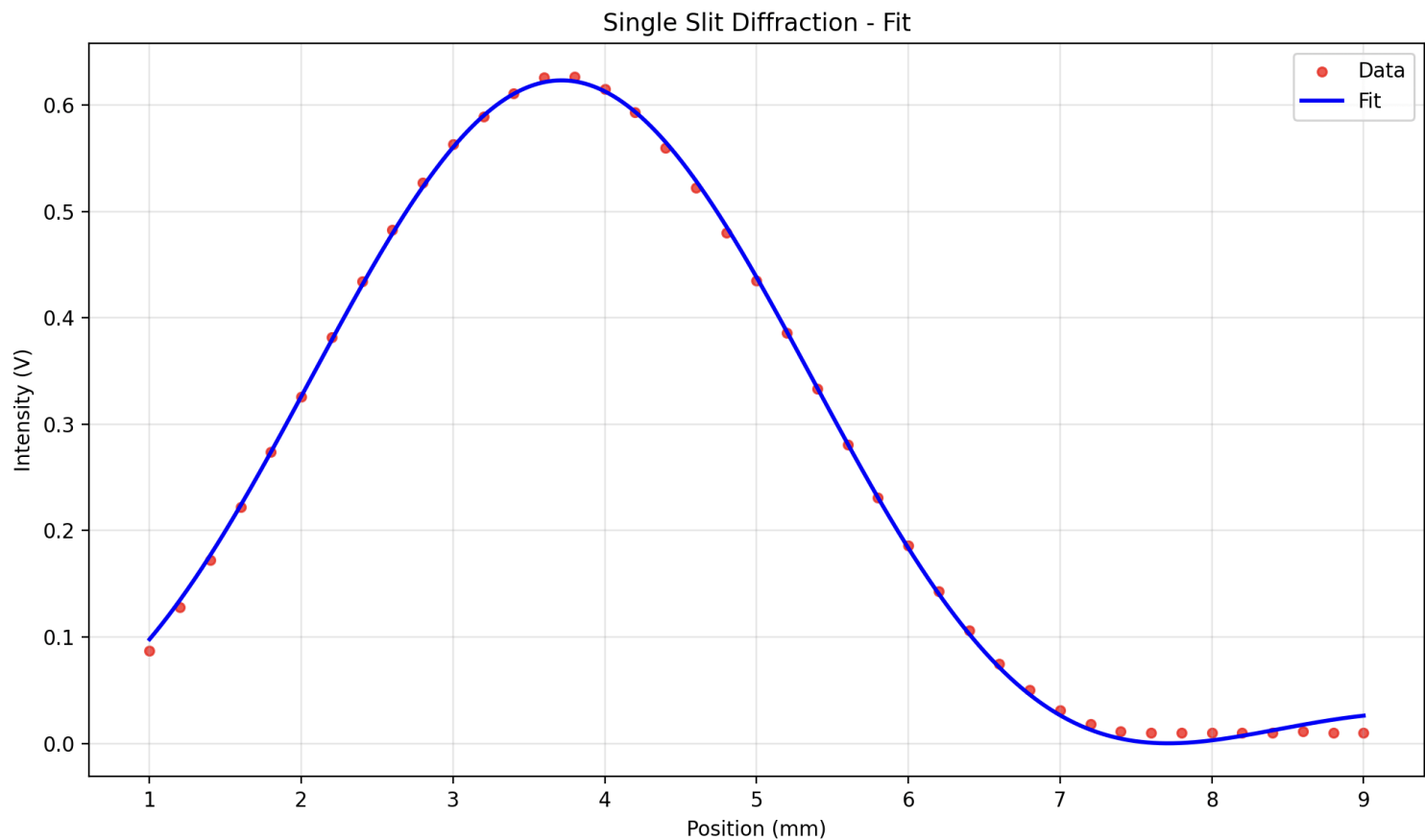


Figure 11 b: Laser single slit prediction and data

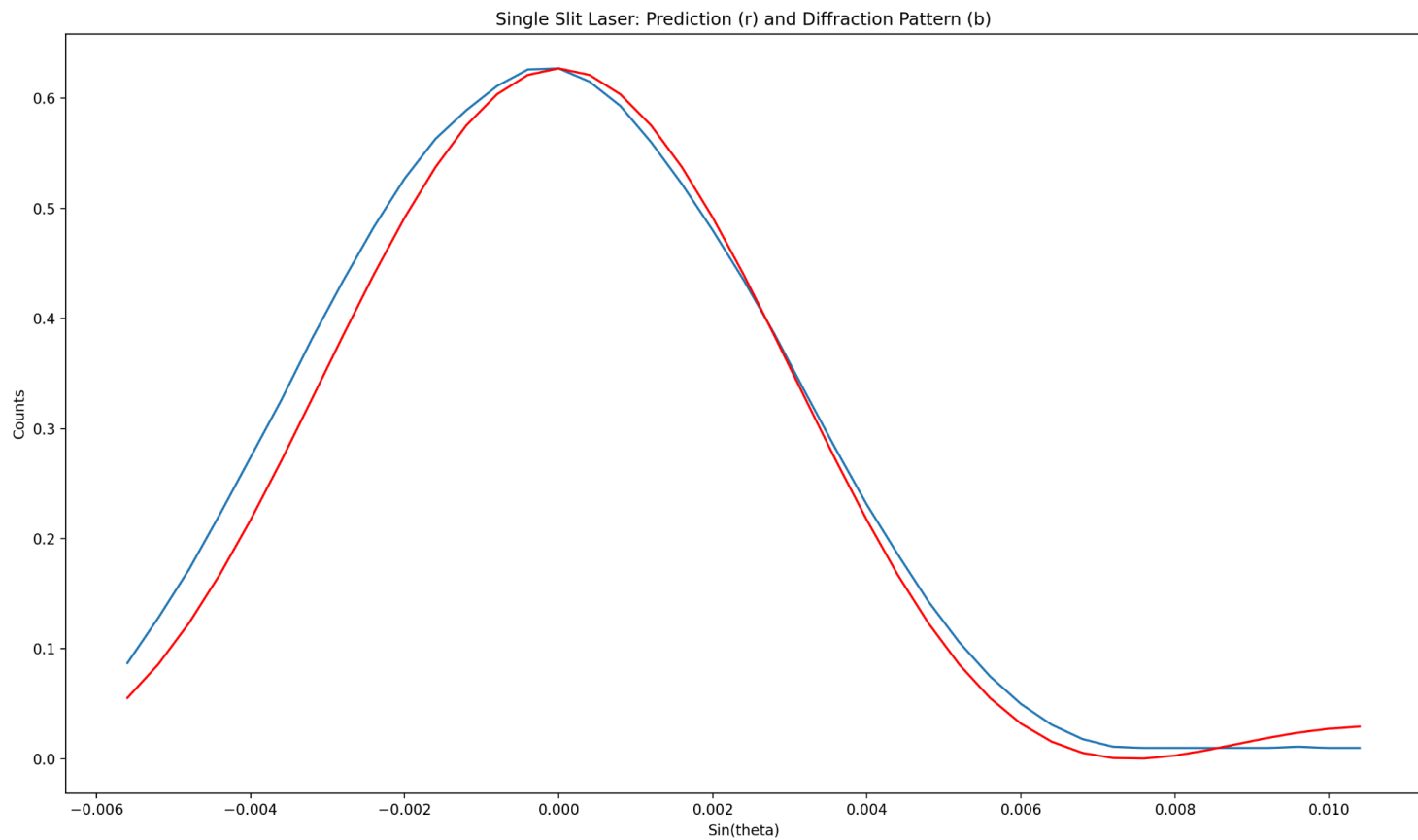


Figure 12 a: laser double slit fit

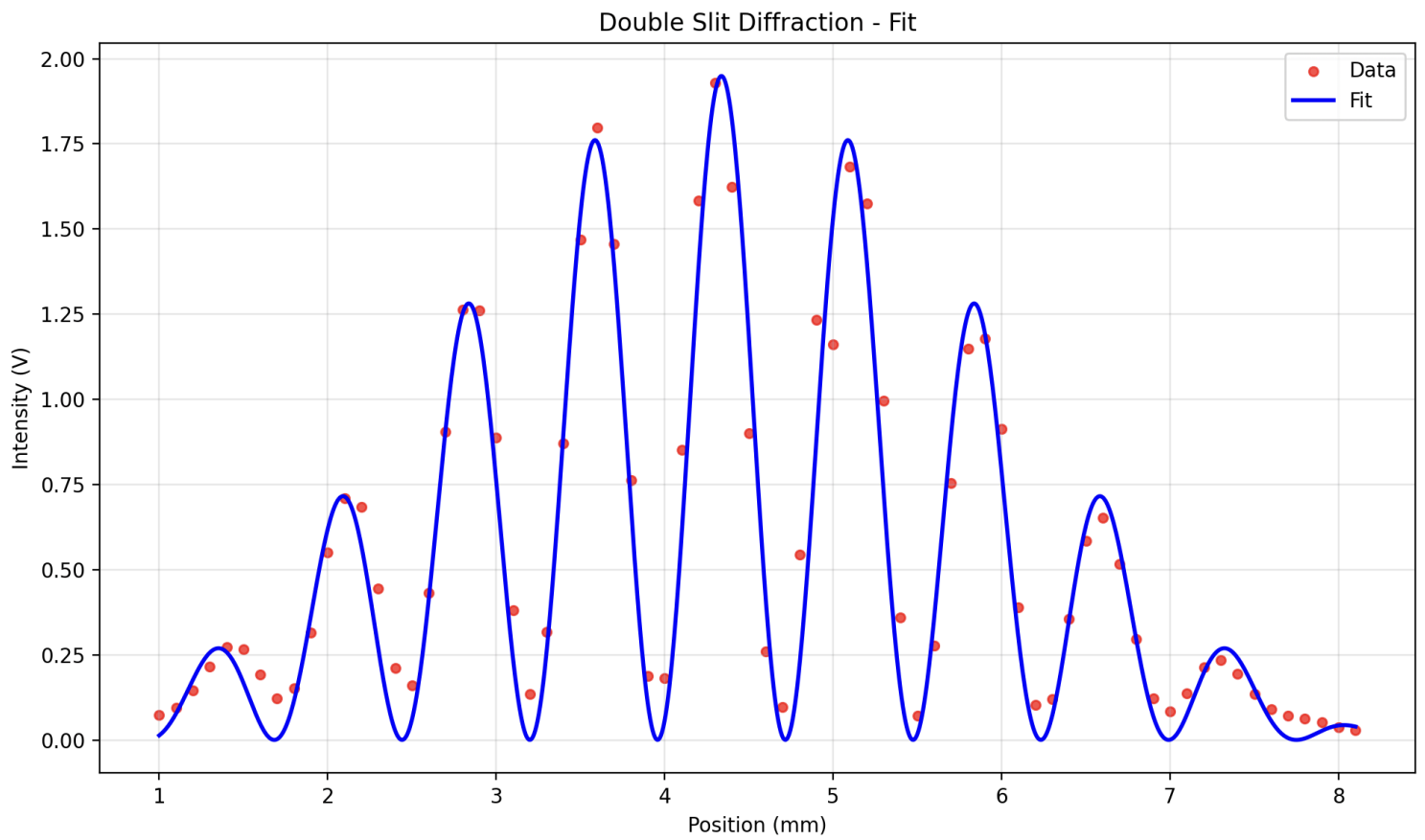


Figure 12 b: laser double slit prediction and data

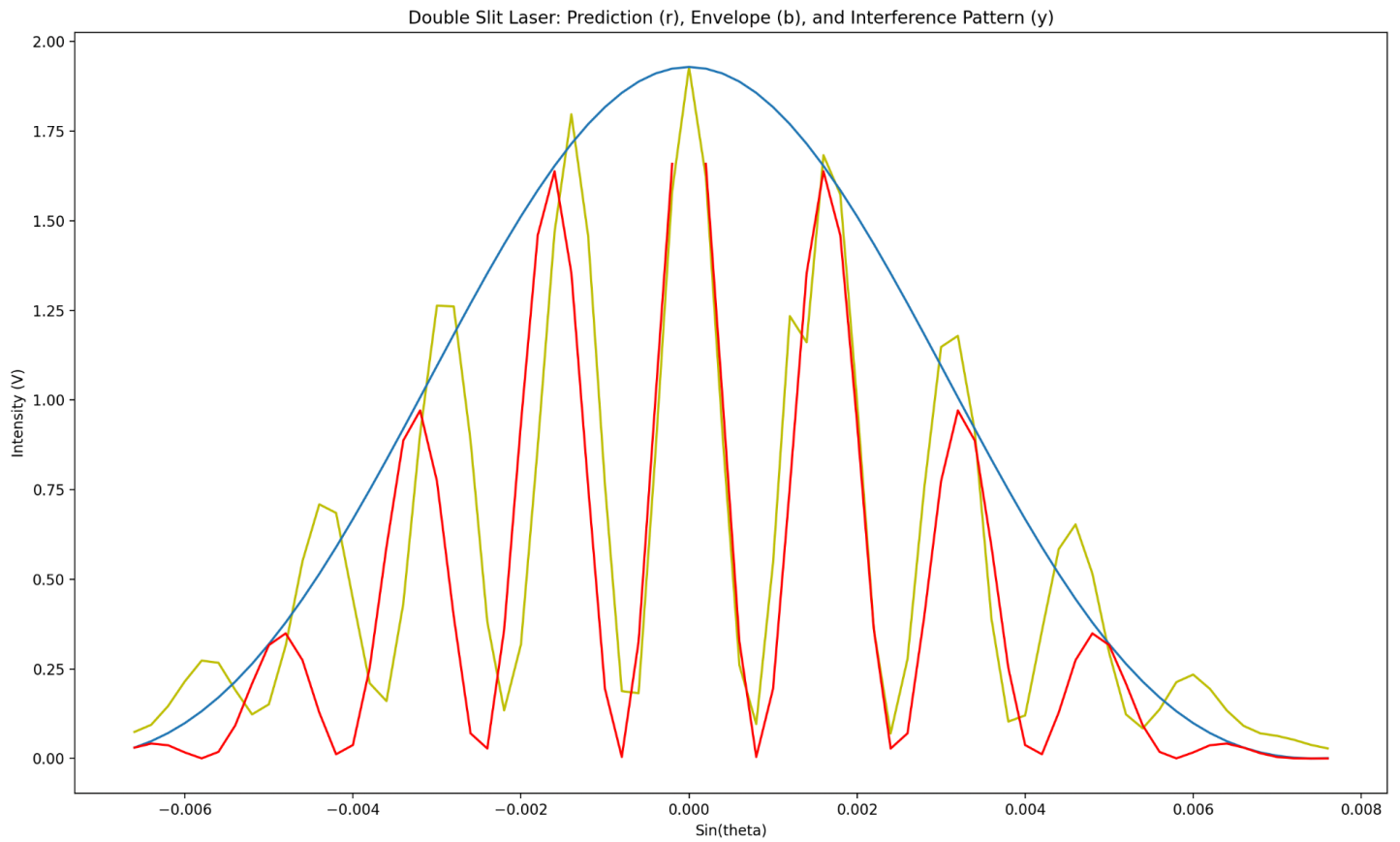


Figure 13: Single Slit Single Photon Diffraction Pattern

Single Slit Single Photon Pattern

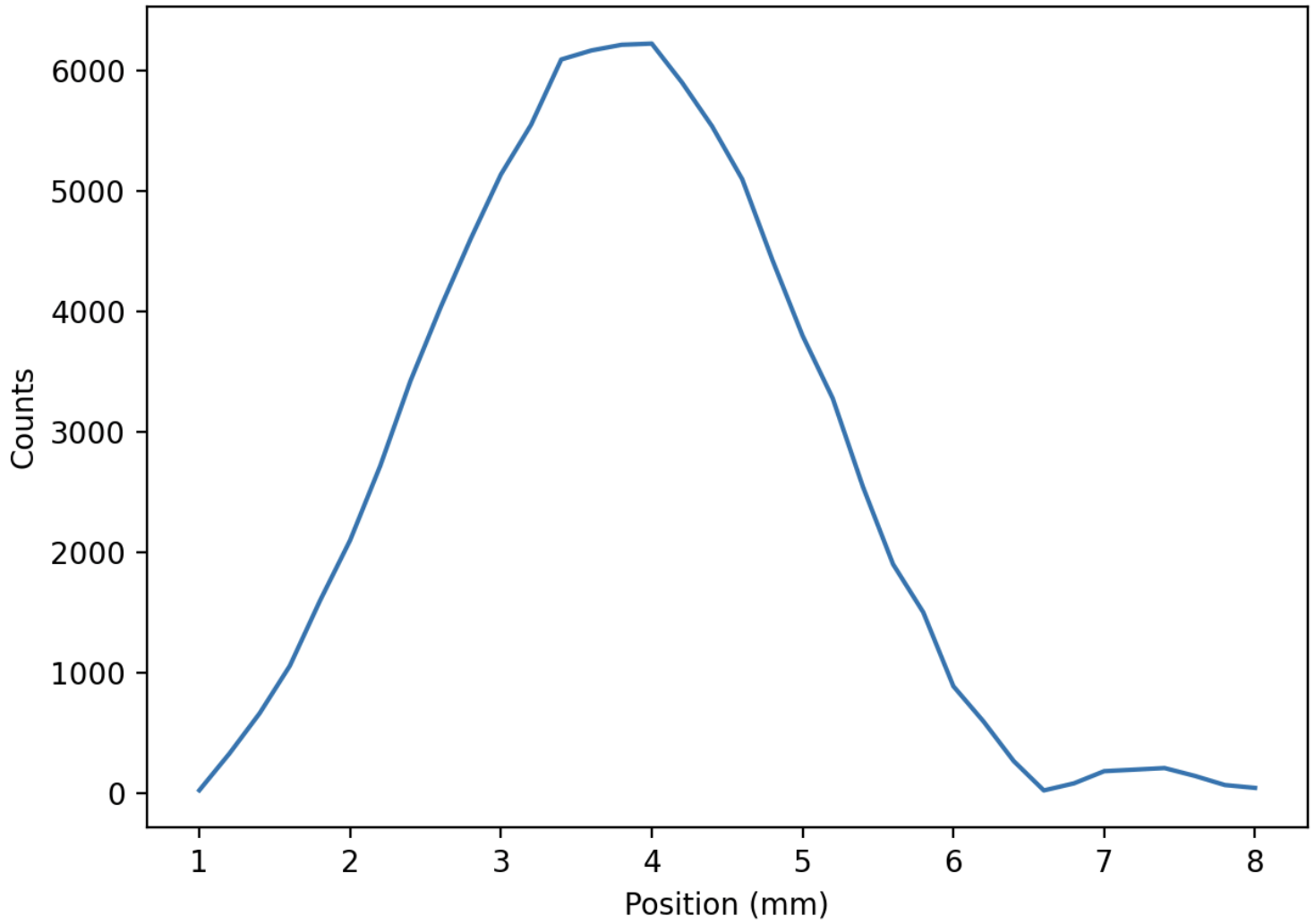


Figure 14: Double Slit Single Photon Interference Pattern

Double Slit Single Photon Pattern

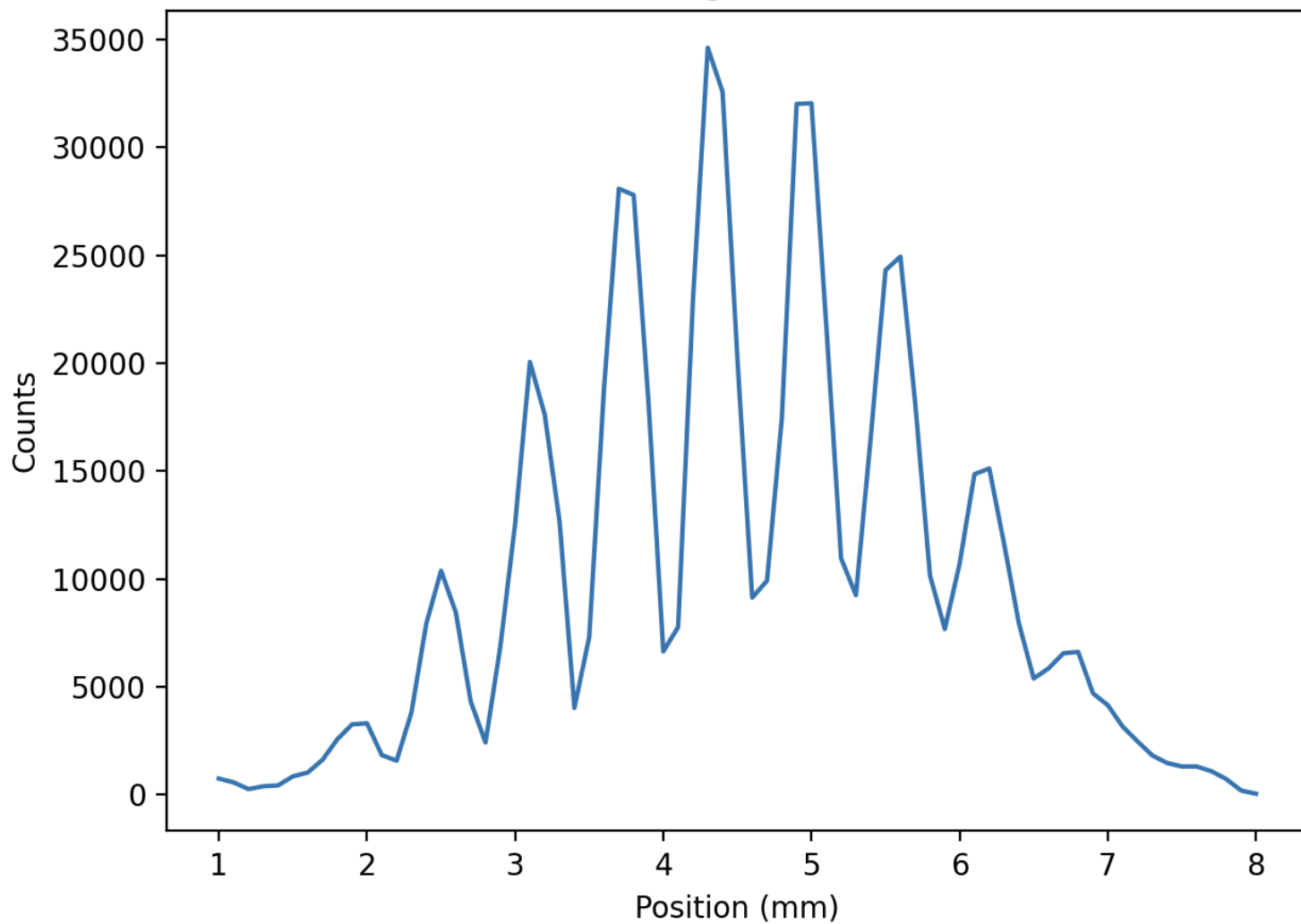


Figure 15 a: Single slit single photon fit

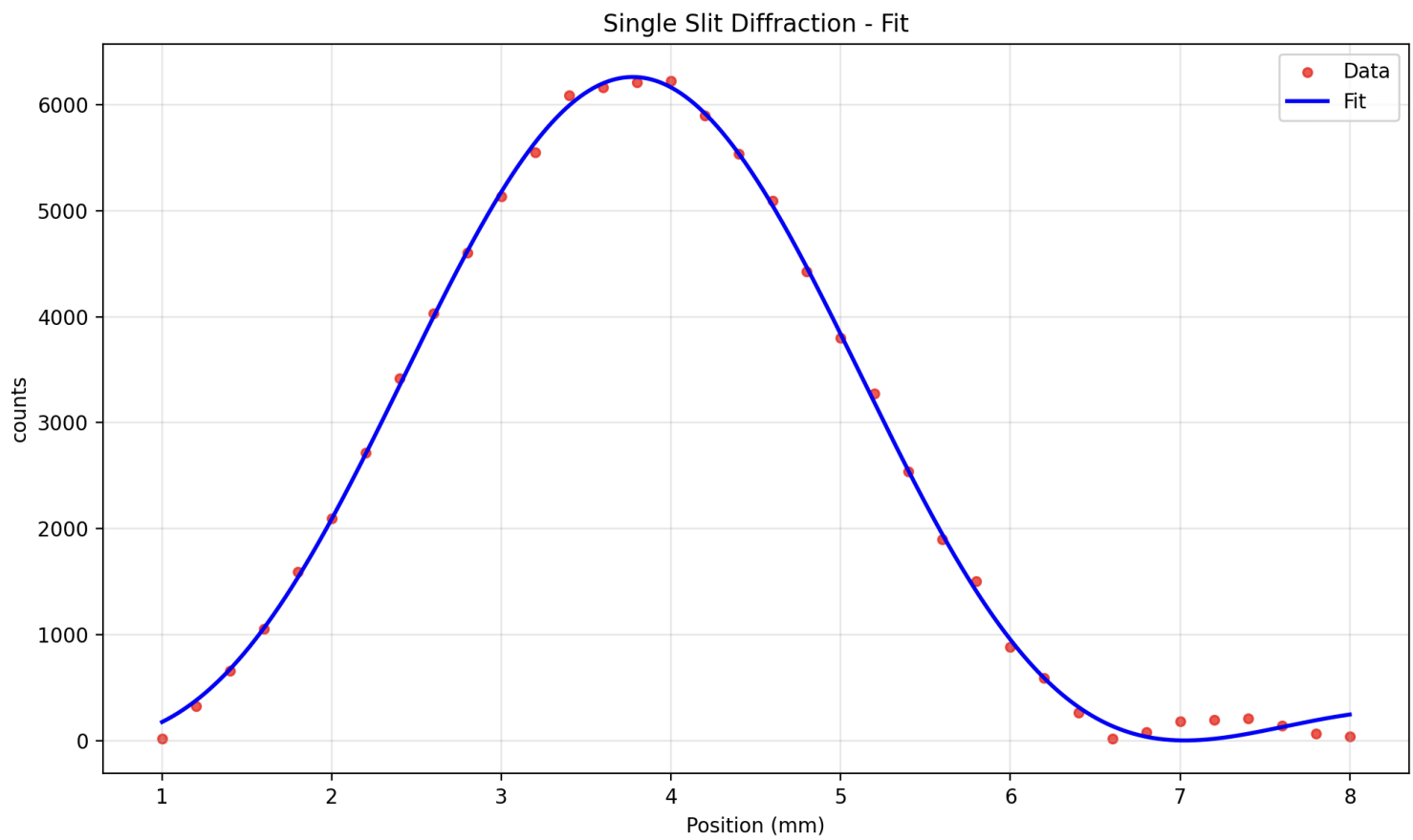


Figure 15 b: Single slit single photon prediction and data

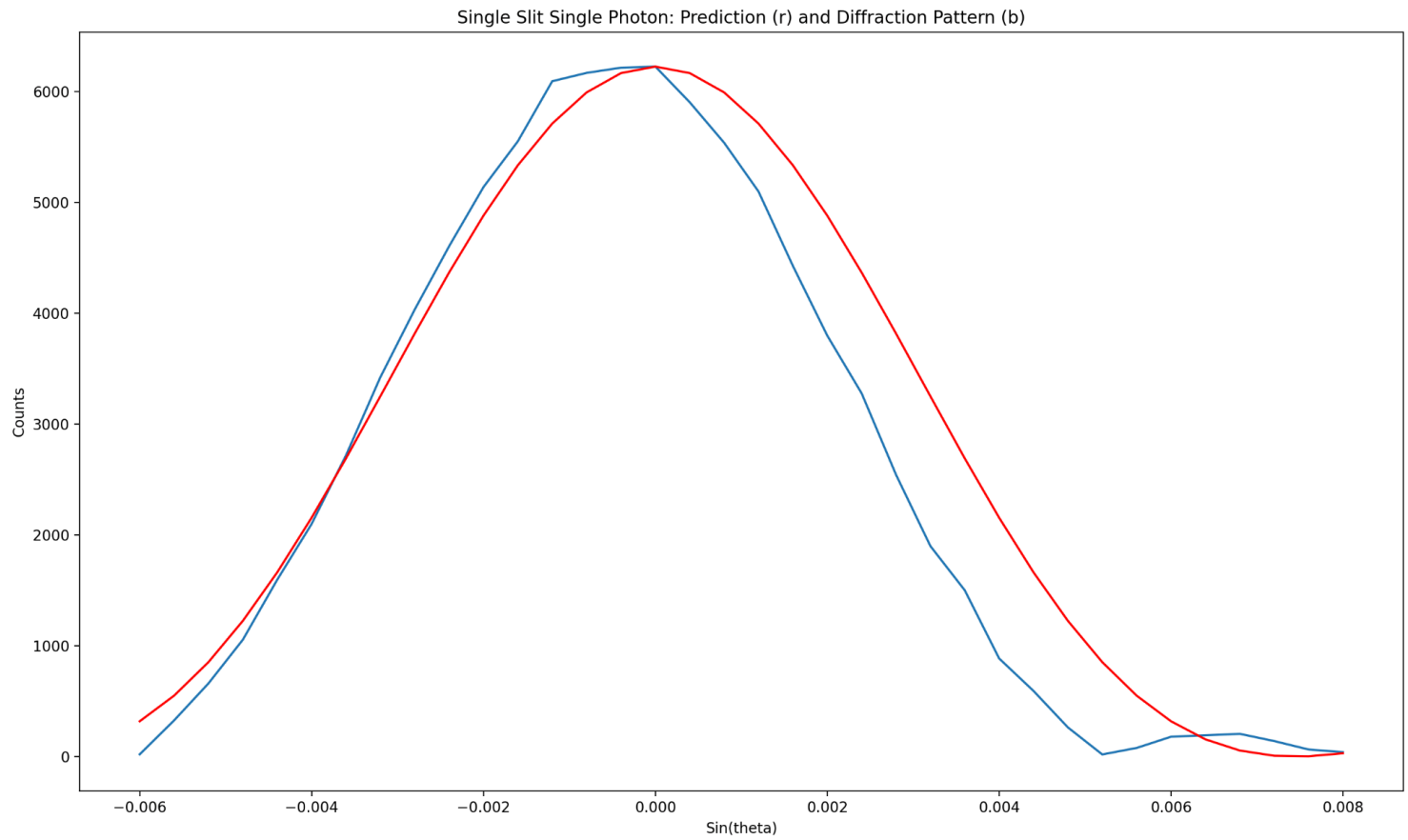


Figure 16 a: Double slit single photon fit

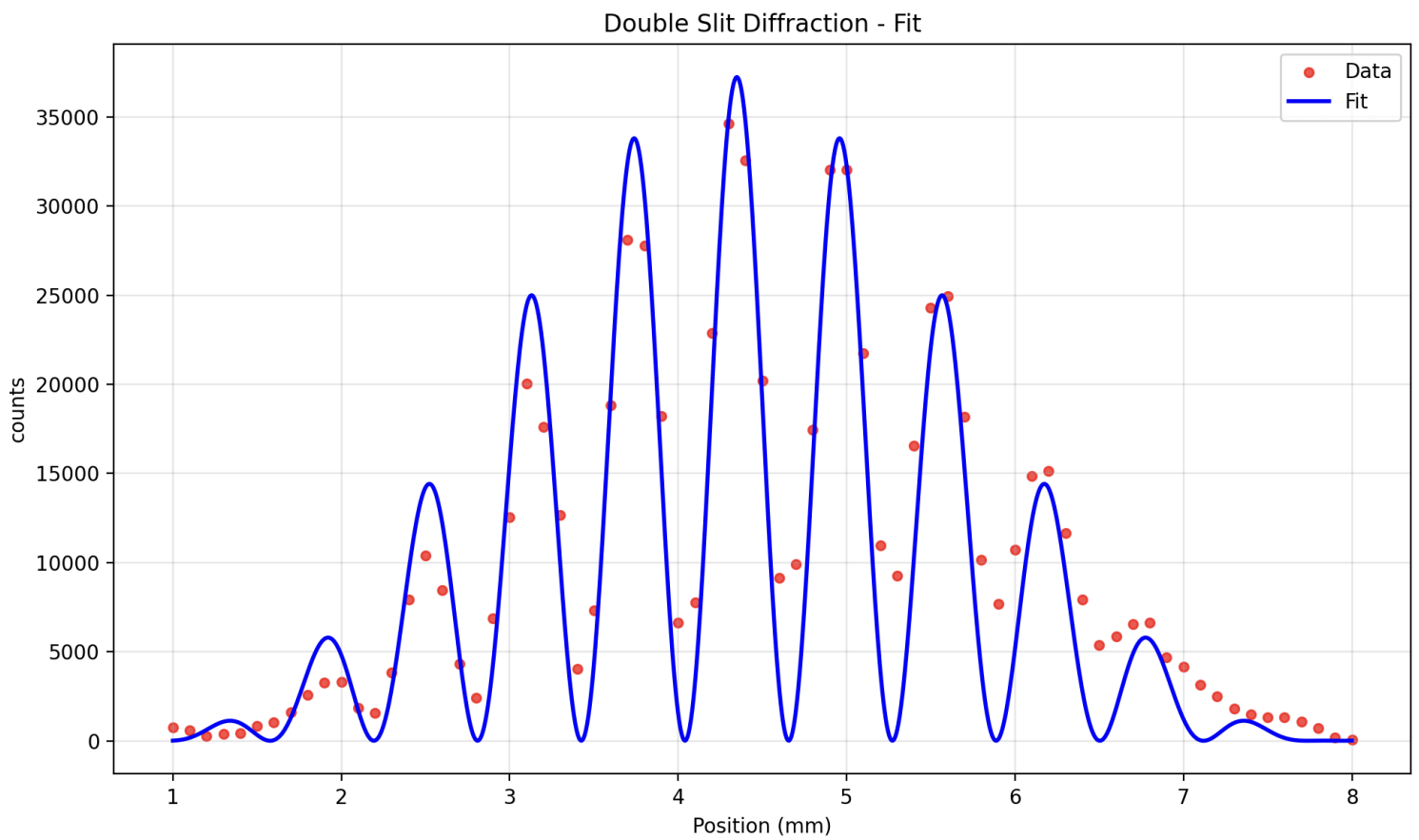
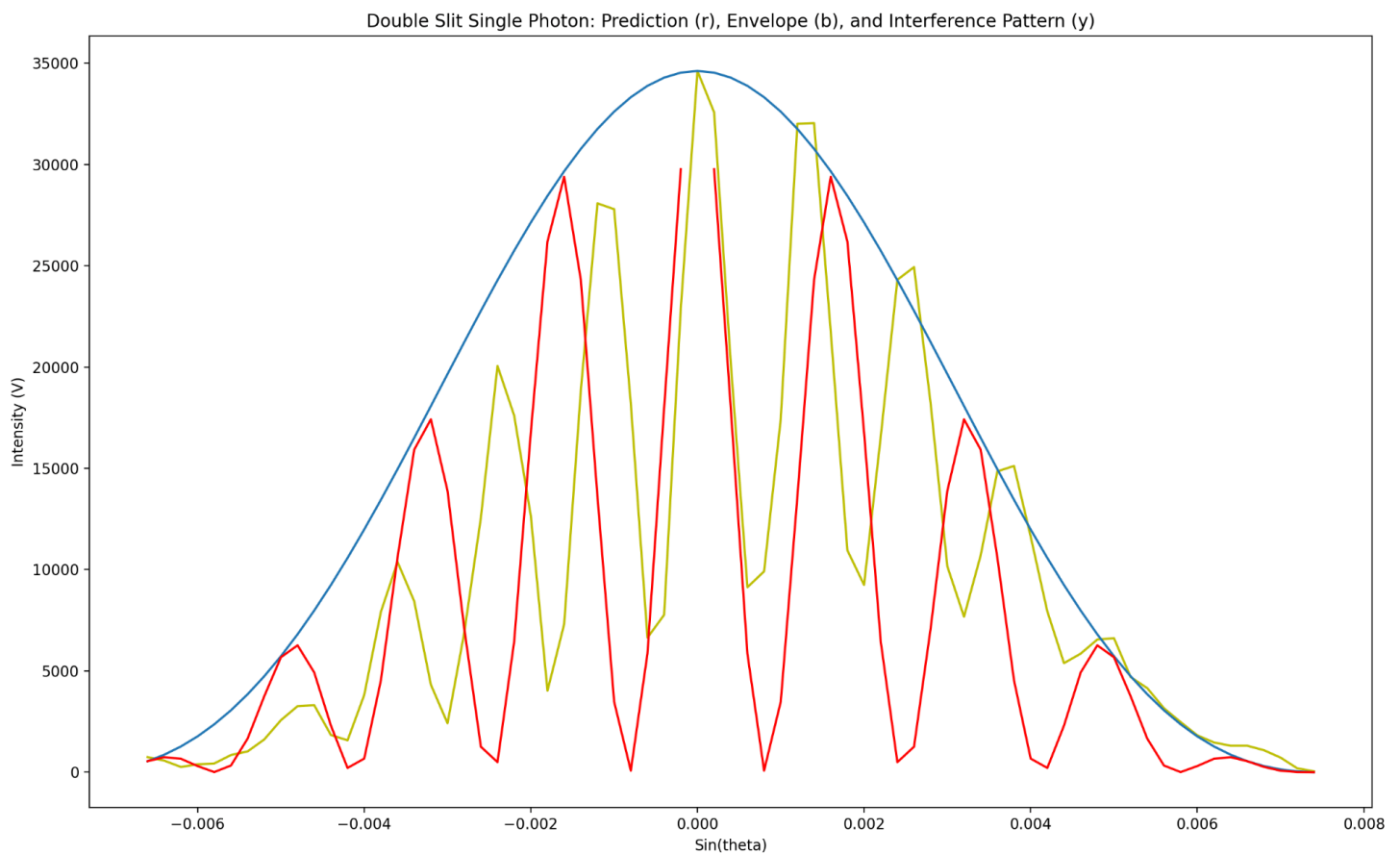


Figure 16 b: Double slit single photon prediction and data



Conclusion:

We were able to observe single slit diffraction for both laser and single photons and interference patterns with double slits. We plotted the data with the predictions using the equations for diffraction and interference and

compared our measurements with theory and we got consistent results. We were able to conclude that diffraction only has an envelope but interference has modulation